

WEEK-4 ASSESSMENT

1. A context switch occurs when a state of process changes from new to ready

1 Point

- ☐ True
- ☒ False

2. A thermal sensor in the CPU checks the temperature of the machine. If it is more than 40°C an interrupt is generated to shutdown the machine. This interrupt is a hardware interrupt.

1 Point

- ☒ True
- ☐ False

3. What is the correct order of the following events

- I. device raises interrupt
- II. CPU checks signal on INT pin and sends INT ACK signal
- III. Interrupt Handler Routine is executed
- IV. Interrupt Controller sends INT signal to CPU
- V. CPU identifies which device has raised the interrupt

- ☐ I,II,III,IV,V
- ☐ V,I,IV,II,III
- ☒ I,IV,II,V,III
- ☐ II,IV,I,V,III

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4. I/O APICs are used to handle interrupt from thermal sensors and internal timers

- ☐ True
- ☒ False

5. Match the following Mnemonics of Interrupt Vector used in x86 to the type of interrupt the are

I. #0F a. Trap
II. #DF b. Fault
III. #SS c. Abort

- ☐ I - a II - b III - c
- ☒ I - a II - c III - b
- ☐ I - b II - c III - a
- ☐ I - c II - b III - a

6. A CPU with four processes executing has an Operating System with a non-preemptive scheduler. What is the total number of scheduling possible for the 4 processes?_____ 1

24

7. More than one System call in an operating system can have the same system call number 1

- ☐ True
- ☒ False

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8. Parameters in system calls are passed using user mode stacks in Linux and by using registers in xv6.

- ☐ True
- ☒ False

9. Read the following statements and point out which are true and false

- I. A context Switch can occur when a process moves from a blocked to ready state
- II. A context Switch can occur when a process moves from a running to zombie state
- III. A context Switch can occur when a process moves from a running to ready state

- ☒ I - False , II - True, III - True
- ☐ I - False , II -False, III - True
- ☐ I - True, II - True, III - True

10. Timer Interrupt Latency should be as high as possible for smooth context switching

- ☐ True
- ☒ False